Gaussian Quadratic Function Report

Polynomial

$$f(q) = aq^2 + bq + c$$

Naive

$$g_{\mathrm{naive}}(X) = Q^2 a + Qb + c$$

$$\mathbb{E}[g_{\mathrm{naive}}(X)] = aq^2 + a\sigma^2 + bq + c$$

$$\mathrm{Var}(g_{\mathrm{naive}}(X)) = \sigma^2 \left(4a^2q^2 + 2a^2\sigma^2 + 4abq + b^2\right)$$

$$\mathrm{MSE}(g_{\mathrm{naive}}) = \sigma^2 \left(4a^2q^2 + 3a^2\sigma^2 + 4abq + b^2\right)$$

$$\mathrm{Bias}(g_{\mathrm{naive}}) = a\sigma^2$$

Unbiased

$$g_{\mathrm{unb}}(X) = Q^2 a + Qb - a\sigma^2 + c$$

$$\mathbb{E}[g_{\mathrm{unb}}(X)] = aq^2 + bq + c$$

$$\mathrm{Var}(g_{\mathrm{unb}}(X)) = \sigma^2 \left(4a^2q^2 + 2a^2\sigma^2 + 4abq + b^2\right)$$

$$\mathrm{MSE}(g_{\mathrm{unb}}) = \sigma^2 \left(4a^2q^2 + 2a^2\sigma^2 + 4abq + b^2\right)$$

Comparison

$$\text{mean gap} = -a\sigma^2$$

$$\text{variance gap} = 0$$

$$\text{MSE gap} = -a^2\sigma^4$$

$$\mathrm{Bias}(g_{\mathrm{naive}})^2 = a^2\sigma^4$$